

Project Design Phase
Problem – Solution Fit Template

Date	25 JULY 2026
Team ID	
Project Name	CAB BOOKING
Maximum Marks	2 Marks

Problem Statement

Customers often experience difficulty in booking cabs quickly due to the lack of a centralized online platform. Traditional cab booking methods involve long waiting times, uncertain fare estimates, limited driver availability, and the inability to track rides in real time. Drivers also face challenges in receiving ride requests efficiently, while administrators struggle to manage bookings, drivers, and customer information manually.

These issues reduce customer satisfaction, increase operational inefficiency, and make transportation services less reliable.

Problem	Impact
Difficulty finding nearby cabs	Increased waiting time
Unclear fare estimation	Lack of pricing transparency
No real-time ride tracking	User uncertainty and safety concerns
Manual booking process	Slow and inefficient service
Limited online payment options	Payment inconvenience
Poor booking management	Reduced customer satisfaction
Manual driver management	Increased administrative workload

Proposed Solution

The **Cab Booking System** provides a complete online platform where customers can search for available cabs, estimate fares, book rides, track drivers in real time, make secure online payments, and manage booking history. Drivers can receive ride requests instantly and update ride status, while administrators can efficiently manage users, drivers, vehicles, and bookings through a centralized dashboard.

Solution Features

- User Registration and Login
- Driver Registration and Verification
- Search Nearby Cabs
- Online Cab Booking

- Fare Estimation
- Real-Time Ride Tracking
- Secure JWT Authentication
- Online Payment Integration
- Booking History
- Admin Dashboard
- Driver Dashboard
- Responsive User Interface

Customer Problem	Proposed Solution	Customer Benefit
Difficulty booking a cab	Online cab booking system	Quick and convenient booking
Long waiting time	Nearby driver matching	Reduced waiting time
Unclear fare estimates	Fare estimation feature	Transparent pricing
No ride tracking	Real-time GPS tracking	Better safety and confidence
Manual payments	Secure online payment gateway	Fast and secure transactions
No booking records	Booking history module	Easy access to previous rides
Poor driver management	Driver dashboard	Efficient ride assignment
Manual administration	Admin dashboard	Simplified system management

Value Proposition

The **Cab Booking System** offers a fast, secure, and reliable transportation solution by connecting passengers, drivers, and administrators through a single web application. It improves ride booking efficiency, enhances customer safety through live tracking, simplifies payment processing, and provides powerful management tools for administrators.

Expected Benefits

For Customers

- Easy online cab booking
- Real-time ride tracking
- Secure online payments
- Transparent fare estimation
- Booking history management
- Improved travel convenience

For Drivers

- Faster ride allocation
- Efficient ride management
- Earnings tracking
- Better communication with customers

For Administrators

- Centralized user management
- Driver verification and monitoring
- Booking management
- Business analytics and reporting
- Improved operational efficiency

Target Customers

- Daily commuters
- Students
- Office employees
- Tourists and travellers
- Drivers
- Cab service administrators
- **1. Customer Segment(s):** Daily commuters, Students, Office employees, Tourists, Drivers, Administrators.
- **2. Jobs-to-be-Done / Problems:** Book cabs quickly, Find nearby drivers, Fare estimation, Real-time ride tracking, Secure online payments, Booking history.
- **3. Triggers:** Urgent travel, Daily commuting, Airport trips, Late-night travel, Heavy traffic, Recommendations.
- **4. Emotions (Before/After):** Before: Frustrated, Worried, Uncertain. After: Happy, Safe, Confident, Satisfied.
- **5. Available Solutions:** Traditional taxis, Auto-rickshaws, Ride-hailing apps, Public transport, Phone booking.
- **6. Customer Constraints:** Poor internet, Budget limitations, Driver availability, Traffic congestion, GPS inaccuracies, Limited payment methods.
- **7. Behaviour:** Compare fares, Search nearby cabs, Track driver location, Pay online, Rate drivers.
- **8. Channels of Behaviour:** Online: Website, Mobile Browser, Email, Social Media. Offline: Taxi stands, Customer Support, Travel Agencies.
- **9. Problem Root Cause:** Manual booking, No centralized platform, Poor communication, No live tracking, Inefficient management.
- **10. Your Solution:** Develop a MERN Stack Cab Booking System with secure JWT authentication, nearby cab search, fare estimation, live GPS tracking, online payment, booking history, driver management, admin dashboard, and responsive UI.

